



The Sun — Our Closest Star

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 1 of 6



Grade: 1st Grade



Duration: 45 minutes



Subject: Earth Science — Sun, Moon & Stars



Materials: Household only



Sun Safety — Read Before Starting!

Never look directly at the sun — not even for a split second. The sun's light is so powerful it can permanently damage your eyes. This lesson uses *shadows* as a safe way to study the sun — we never look at the sun itself. Please review this rule with your child before going outside.

- Always keep your back to the sun when doing shadow observations
- If you feel the urge to look up — look at the ground instead
- Wear sunscreen and a hat on sunny days



The sun, Earth, and its shadow — a system that creates our days and nights



NGSS Standard: 1-ESS1-1 — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

Learning Objectives

By the end of this lesson, students will be able to:

- *Describe* the observable pattern of the sun's daily motion across the sky
- *Explain* why the sun appears to move from east to west each day
- *Connect* shadow length and direction to the sun's position in the sky



The Big Science Idea

The sun appears to move across the sky each day — rising in the east every morning and setting in the west every evening. But here's the surprising truth: *the sun isn't moving!* It's *Earth* that's spinning.

Earth rotates (spins on its axis) once every 24 hours. As we spin, the sun appears to travel across the sky — just like when you spin around in a circle and the trees around you seem to move even though they're standing still. We call this apparent motion.

Because of the sun's position in the sky, shadows change throughout the day. In the morning and afternoon, shadows are long. At midday (when the sun is highest), shadows are shortest.



Key vocabulary: sun, rotate, orbit, shadow, east, west, midday, apparent motion



Materials



What You'll Need

- A sunny day (or a sunny window)
- A stick, pencil, or small object to cast a shadow (a "gnomon")
- A large piece of white paper or poster board
- Tape (to secure paper outside if needed)
- A marker or pencil to trace shadows
- A ruler or measuring tape
- A watch or clock
- This worksheet (printed)
- Crayons

No sunny day? Use a lamp at a fixed position indoors as a "sun" and a small object to cast shadows at different lamp angles to simulate morning, midday, and afternoon.



Let's Investigate — Shadow Tracking!

Step 1 — Set Up (Morning, before 10am)

Place your paper on a flat, sunny spot outdoors (or by a sunny window). Stand a stick or pencil upright in the center — hold it straight or tape it if needed. *Without looking at the sun*, note that the sun is low in the sky. Trace the shadow on the paper and write the time. Measure the shadow's length and note which direction it points.

Step 2 — Midday Observation (Around noon)

Return to the same spot at midday. Trace the shadow again using a different color. Measure its length. Notice: is it longer or shorter than the morning shadow? Which direction does it point now?

Step 3 — Afternoon Observation (2–4pm)

Return once more in the afternoon. Trace the shadow a third time in yet another color. Measure and compare. The shadow should now point in a different direction from the morning shadow.

Step 4 — Record & Discuss

Fill in the Shadow Tracking Record (on the Student Worksheet page). Look at all three shadow tracings together and answer: What changed? What pattern do you notice?



What to expect: Morning shadows point roughly west (sun is in the east). Midday shadows are shortest and point roughly north (sun is highest, due south). Afternoon shadows point roughly east (sun is in the west). The shadow sweeps in a wide arc — this is Earth rotating, not the sun moving!



Discussion Questions



Talk about these together:

- What happened to the shadow's length from morning to midday to afternoon?
- What happened to the direction the shadow pointed?
- If a shadow is very short, where is the sun in the sky?
- If a shadow is very long, where is the sun?



Big Question: "The sun looks like it moves across the sky. Is it actually moving?"

No! The sun stays relatively fixed while Earth spins. Earth rotates once every 24 hours, so from our perspective on the spinning Earth, the sun appears to move from east to west. It's like watching trees go by from a moving car — the trees aren't moving, you are. Ancient people thought the sun moved around Earth (the geocentric model), but we now know Earth orbits the sun (the heliocentric model).



Extensions

-  **Shadow Clock:** Try using your shadow tracings to tell time! Ancient people built sundials using this exact idea.
-  **Compass Connection:** If you have a compass, check which direction the midday shadow points. It should point roughly north (in the Northern Hemisphere) — this is how explorers used the sun for navigation.
-  **Seasonal Compare:** Do this activity again in 3 months. Are the shadows the same? (Hint: they won't be — connect to Lesson 4!)
-  **Read Aloud:** "*The Sun: Our Nearest Star*" by Franklyn Branley



Parent/Teacher Notes

- **Safety first:** Reinforce the no-looking-at-the-sun rule before and during the activity. Make it a habit, not a warning.
- **Timing flexibility:** You don't have to do all three observations in one day. You can do morning one day and midday and afternoon another — just keep the paper setup in exactly the same spot.
- **Common misconception:** Many children (and adults!) believe the sun moves around Earth. This is a great chance to introduce heliocentric thinking without overwhelming — the key idea is just "Earth spins, so the sun seems to move."
- **Grade 1 level:** Don't worry about precise compass directions. The pattern to internalize is: shadow moves throughout the day, shortest at midday.
- **If no sun:** The lamp simulation works well. Hold the lamp at a low angle (morning), high angle (midday), and low angle from the other side (afternoon). Let the child hold the object and trace the shadows.

Student Worksheet — Shadow Tracker

Unit 1, Lesson 1 — The Sun

Name: _____ Date: _____

Shadow Tracking Record

Fill in the table as you make each observation. Use a different color for each time of day!

Time of Day	Approximate Time	Shadow Length	Direction Shadow Points	Color Used
Morning				 Red
Midday				 Blue
Afternoon				 Green

Draw Your Shadows

Draw a bird's-eye view (looking down from above) of your shadow tracings. Show all three shadows coming from the same center point. Label each one!

Draw your morning, midday, and afternoon shadows here

Bird's-eye view of shadow tracings — label morning , midday , afternoon 

Answer These Questions

- Which shadow was the *shortest*? _____ Why do you think so? _____
- Did the shadow move? (circle) **YES** / **NO** — What made it move? _____
- True or false: The sun moves across the sky. (circle) **TRUE** / **FALSE**

4. What is *actually* moving? _____

★ Think Like a Scientist

A scientist makes a *prediction* before an experiment. What do you predict will happen to the shadows if you do this activity again *at night* with a flashlight pretending to be the sun?



PARENT ANSWER KEY — Detach before giving worksheet to students



Shadow Tracking — Expected Results

- **Morning:** Shadow is long, points roughly west (away from the rising sun in the east)
- **Midday:** Shadow is shortest, points roughly north (sun is highest in the southern sky)
- **Afternoon:** Shadow is long again, points roughly east (sun is sinking in the west)



Worksheet Answers

1. The *midday* shadow is shortest. Because the sun is highest in the sky at midday, so light comes from almost straight above and makes a short shadow.
2. Yes, the shadow moved. The shadow moves because Earth is rotating (spinning), which makes the sun appear to be in a different position in the sky.
3. **False.** The sun does not actually move — it only appears to move.
4. *Earth* is actually moving (rotating/spinning).



Think Like a Scientist — Sample Answer

With a flashlight at night, you could shine it from one side (morning), above (midday), and the other side (afternoon) and get similar shadow patterns. The flashlight represents the sun. The pattern would be the same because the light source position determines shadow direction.



What Success Looks Like

- Child can describe that shadows changed direction and length throughout the day
- Child understands that the shortest shadow occurred at midday (when the sun was highest)
- Child can state that *Earth* rotates, making the sun appear to move
- Child followed the sun safety rule throughout the activity